

# The Boris Pusher, and ODEs across Physics



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# Today

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- **The Boris pusher.** The standard integrator for charged particles, and why it conserves energy.
- **Orbits.** How NASA chose an integrator for spacecraft, and why it still uses it.
- **Molecular dynamics.** Velocity Verlet on a gas of atoms, and three things it shows.
- **Cosmology.** Why the microwave background is an ODE problem.

# Part I · The Boris pusher



Charged particles

# A charged particle in a magnetic field



$$\frac{d\mathbf{x}}{dt} = \mathbf{v}, \quad \frac{d\mathbf{v}}{dt} = \frac{q}{m}(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

- With  $\mathbf{E} = 0$  the magnetic force is always perpendicular to  $\mathbf{v}$ : it does no work, and  $|\mathbf{v}|$  never changes.
- In a uniform  $\mathbf{B}$  the particle circles at the cyclotron frequency  $\omega_c = qB/m$ .
- A good method should keep  $|\mathbf{v}|$  exactly, over millions of gyrations.

# Leapfrog for the Lorentz force



Stagger as in Lecture 10: positions at whole steps, velocities at half steps.

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \Delta t \mathbf{v}_{n+1/2}$$
$$\frac{\mathbf{v}_{n+1/2} - \mathbf{v}_{n-1/2}}{\Delta t} = \frac{q}{m} \left( \mathbf{E}_n + \frac{\mathbf{v}_{n+1/2} + \mathbf{v}_{n-1/2}}{2} \times \mathbf{B}_n \right)$$

- The force depends on the velocity, which lives at half steps, so it is averaged over the step.
- The unknown  $\mathbf{v}_{n+1/2}$  appears on both sides: the update looks implicit.

# Boris's trick (1970)

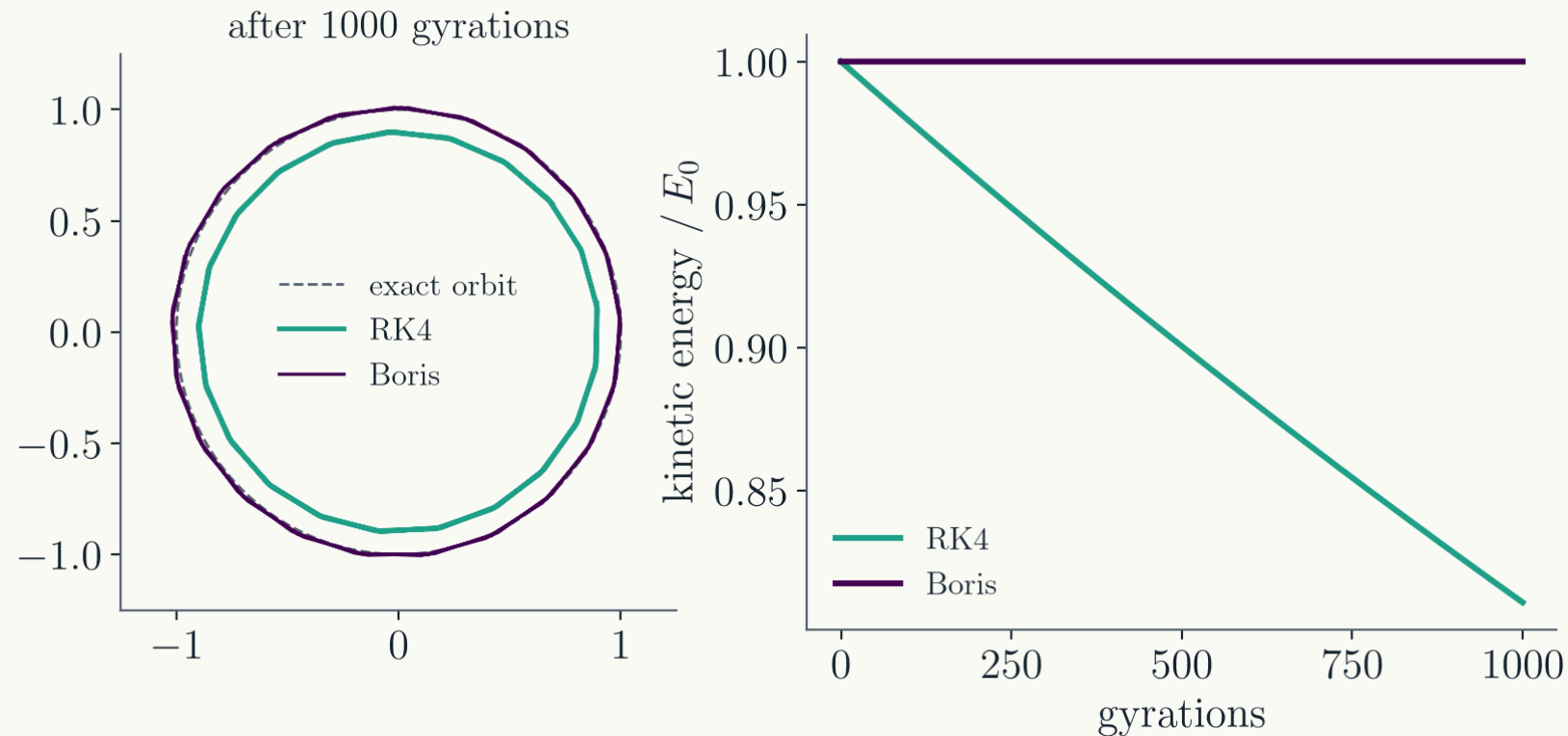


Half an electric kick, a rotation about  $\mathbf{B}$ , and another half kick:

$$\begin{aligned}\mathbf{v}^- &= \mathbf{v}_{n-1/2} + \frac{q\mathbf{E}}{m} \frac{\Delta t}{2} & \mathbf{t} &= \frac{q\mathbf{B}}{m} \frac{\Delta t}{2}, & \mathbf{s} &= \frac{2\mathbf{t}}{1 + t^2} \\ \mathbf{v}^+ &= \mathbf{v}^- + (\mathbf{v}^- + \mathbf{v}^- \times \mathbf{t}) \times \mathbf{s} & \mathbf{v}_{n+1/2} &= \mathbf{v}^+ + \frac{q\mathbf{E}}{m} \frac{\Delta t}{2}\end{aligned}$$

- Two cross products solve the "implicit" equation exactly: no iteration, no matrix.
- The middle step is an exact rotation, so  $|\mathbf{v}^+| = |\mathbf{v}^-|$ : with  $\mathbf{E} = 0$  the energy is conserved to rounding.

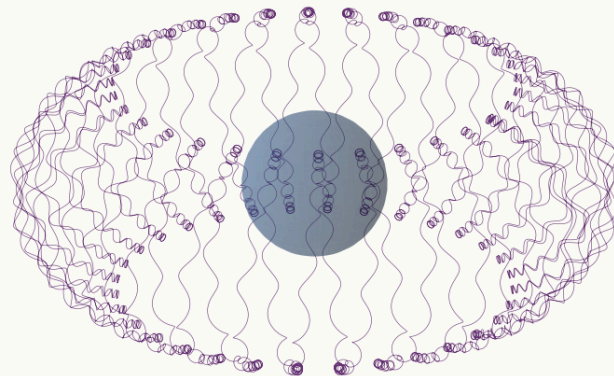
# Boris against RK4, uniform field



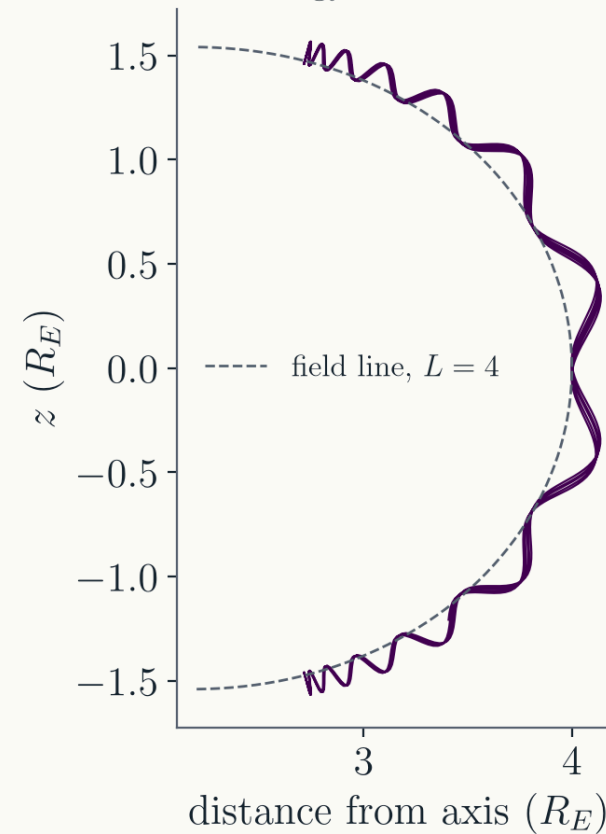
- $\omega_c \Delta t = 0.3$ , about 21 steps per gyration, 1000 gyrations.
- RK4 loses a factor  $1 - (\omega_c \Delta t)^6 / 72$  of energy per step (Lecture 10): 19% by the end. Boris:  $|\mathbf{v}|$  exact to  $10^{-13}$ .

# A proton in the Van Allen belt

75 s: one drift around the Earth



first 6 s: gyration and bounce



10 MeV proton,  $4 R_E$  from Earth's center

gyration

bounce

drift around the Earth

period

0.14 s

2.3 s

1.2 min

# Three timescales, one integrator



- The step is set by the fastest motion, the gyration. The physics of interest (the drift that forms the belt) is 500 times slower.
- Over the drift, Boris conserves the energy to  $10^{-14}$ . RK4, with the same step, loses 98% of it.
- This is why plasma codes push every particle with Boris.

*Structure preservation matters most when the interesting timescale is far longer than the step.*

# Particle-in-cell simulations

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Relativistic kinetic turbulence simulations ( $\sigma=10$ ): density

Joonas Nättiä



Watch on

Relativistic plasma turbulence (J. Nättiä): [youtube.com/watch?v=Gj9mwAww3TM](https://youtube.com/watch?v=Gj9mwAww3TM)

Billions of particles pushed by Boris (ODEs), and the fields on a grid, evolved by Maxwell's equations (PDEs): coming later in the course.

# Research from our group



Space for new research results (1)

# Research from our group



Space for new research results (2)

# Part II · How NASA integrates orbits



Choosing an integrator in 1963

# NASA's comparison, 1963

TABLE 1  
Comparison Criteria

| Method of Numerical Integration                                    | Truncation Error | Ease of Changing Step-Size                  | Speed     | Stability              | Roundoff Error Accumulation |
|--------------------------------------------------------------------|------------------|---------------------------------------------|-----------|------------------------|-----------------------------|
| <b>Single Step Methods</b>                                         |                  |                                             |           |                        |                             |
| Runge-Kutta                                                        | $h^5$            | *                                           | Slow      | Stable                 | Satisfactory                |
| Runge-Kutta Gill                                                   | $h^5$            | *                                           | Slow      | Stable                 | Satisfactory                |
| Bowie                                                              | $h^3$            | Trivial (step-size varied by error control) | Fast      | Stable                 | Satisfactory                |
| <b>Fourth Order Multistep Predictor-corrector</b>                  |                  |                                             |           |                        |                             |
| Milne                                                              | $h^5$            | Excellent                                   | Very fast | Unstable               | Poor                        |
| Adams-Moulton                                                      | $h^5$            | Excellent                                   | Very fast | Unconditionally stable | Satisfactory                |
| <b>Higher Order Multistep</b>                                      |                  |                                             |           |                        |                             |
| Adams Backward Difference                                          | Arbitrary        | Good                                        | Very fast | Moderately stable      | Satisfactory                |
| Gauss-Jackson**                                                    | Arbitrary        | Awkward and expensive                       | Fast      | Stable                 | Excellent                   |
| Obrechhoff                                                         | $h^7$            | Excellent                                   | ***       | Stable                 | Satisfactory                |
| <b>Special Second Order Equations [<math>y'' = f(t, y)</math>]</b> |                  |                                             |           |                        |                             |
| Special Runge-Kutta                                                | $h^5$            | *                                           | Slow      | Stable                 | Satisfactory                |
| Milne-Stormer                                                      | $h^6$            | Excellent                                   | Very fast | Moderately stable      | Poor                        |

\*R-K (single step) trivial to change steps, very difficult to determine proper size.  
 \*\*Gauss-Jackson is for second order equations.  
 \*\*\*Speed of Obrechhoff depends on complexity of the higher order derivatives required; it could be very fast.

NASA Space Flight Handbook, vol. 1 (1963), Chapter IV.C: the methods of the last five lectures, rated for computing trajectories.

Runge-Kutta: easy to start, slow. Multistep: fast.

"Unconditionally stable" here means no growing parasitic roots at small  $h$ , not the stiff stability of Lecture 10.

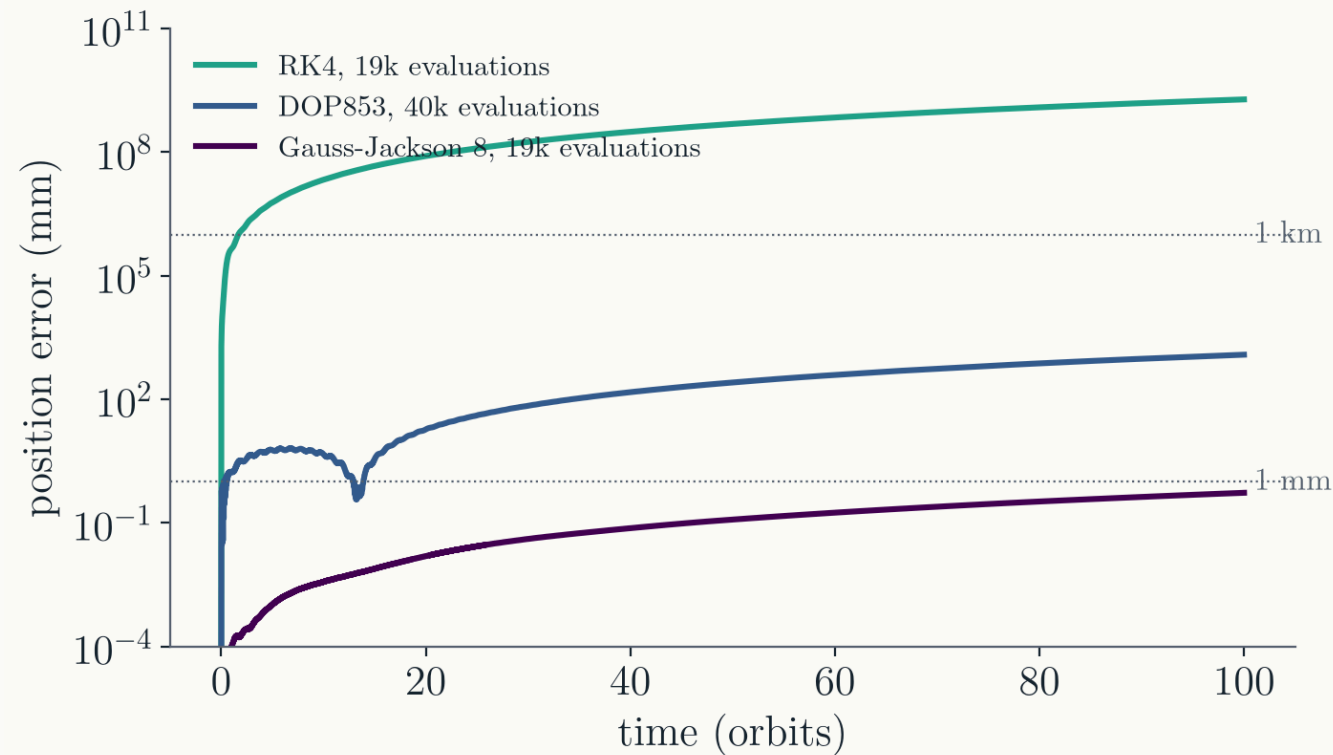
Their pick for orbits: Gauss-Jackson.

# Gauss-Jackson



- A multistep predictor-corrector method (Lecture 10), built for second-order equations  $\ddot{\mathbf{r}} = \mathbf{f}(t, \mathbf{r})$ : it integrates twice without converting to a first-order system.
- Eighth order, fixed step, and about one or two evaluations of the force per step.
- Needs a start-up (a Runge-Kutta run for the first points), like every multistep method.
- Still used for satellite orbit propagation (Berry & Healy 2004).

# 100 orbits, measured



A circular orbit 7000 km from Earth's center (period 97 minutes), error after 100 orbits:

|                      |         |
|----------------------|---------|
| RK4, 20k evaluations | 1900 km |
| DOP853, 40k          | 1.2 m   |
| DOP853, 97k          | 0.7 mm  |
| Gauss-Jackson 8, 20k | 0.5 mm  |

High order, one or two evaluations per step, and a smooth periodic problem: the best case for a fixed-step multistep method.

# Part III · Molecular dynamics



Newton's equations for many atoms

# Molecular dynamics

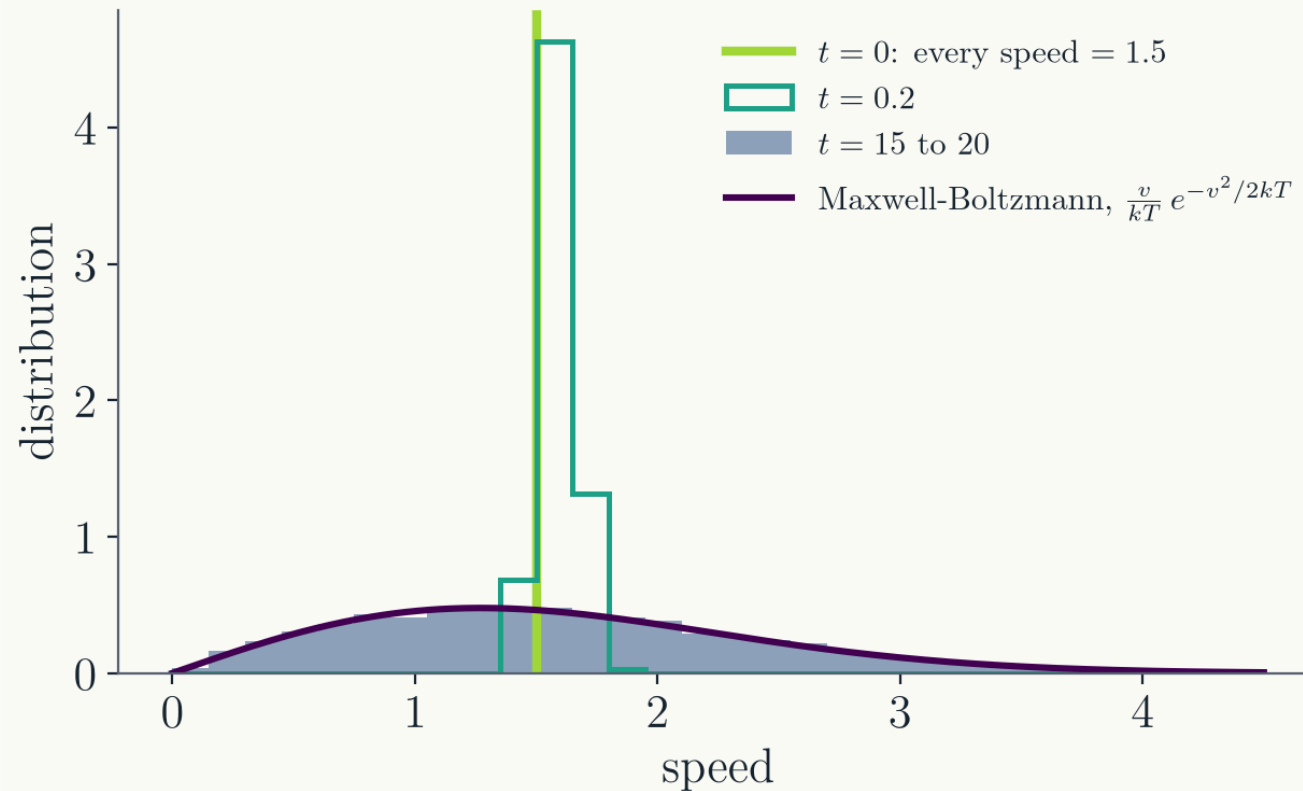


$N$  atoms, Newton's second law, and a pair potential. For noble-gas atoms, Lennard-Jones:

$$m \ddot{\mathbf{r}}_i = - \sum_{j \neq i} \nabla_i V(r_{ij}), \quad V(r) = 4\epsilon \left[ \left( \frac{\sigma}{r} \right)^{12} - \left( \frac{\sigma}{r} \right)^6 \right]$$

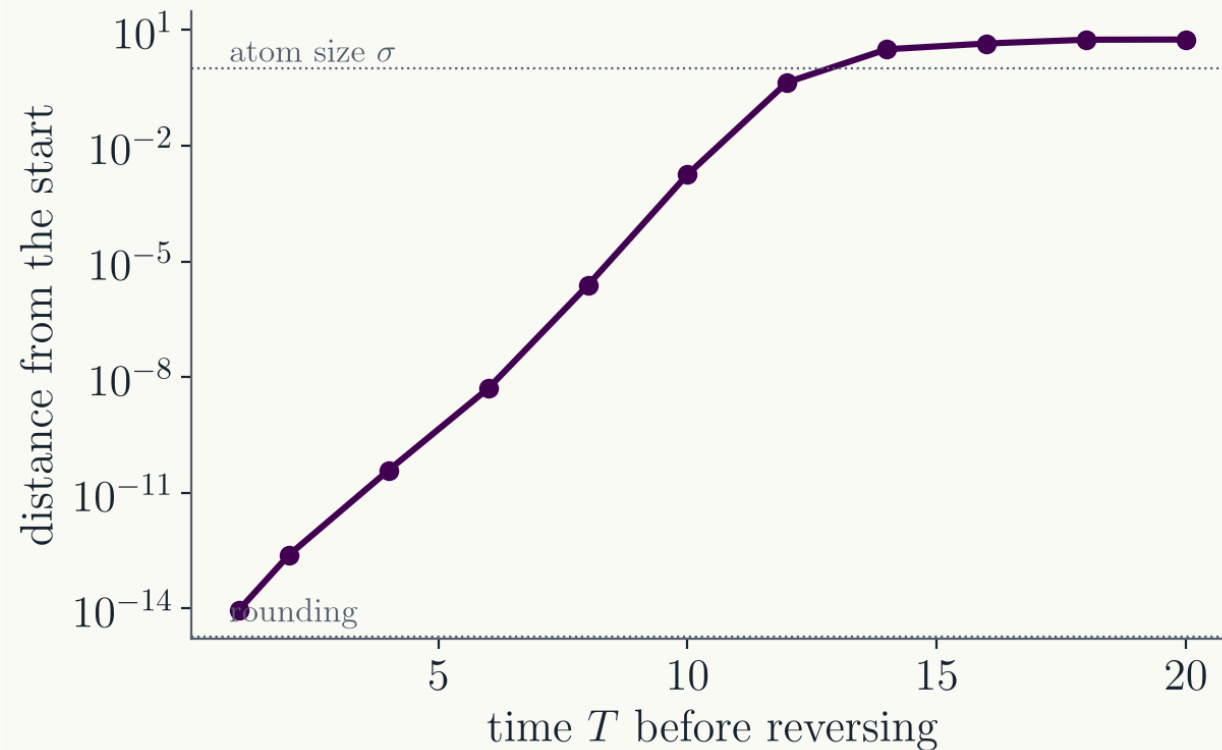
- Integrated with velocity Verlet (Lecture 10): one force evaluation per step, and bounded energy error over millions of steps.
- The cost is in the forces: all pairs is  $O(N^2)$ . Cutting the potential off at a few  $\sigma$  and keeping neighbor lists makes it  $O(N)$ .
- Used for liquids, materials, proteins and drug binding.

# Collisions make Maxwell-Boltzmann



- 400 atoms, every one starting at the same speed, in random directions.
- Nothing statistical is put in: Newton's equations and collisions produce the Maxwell-Boltzmann distribution, at the temperature set by the energy.

# Running the gas backwards

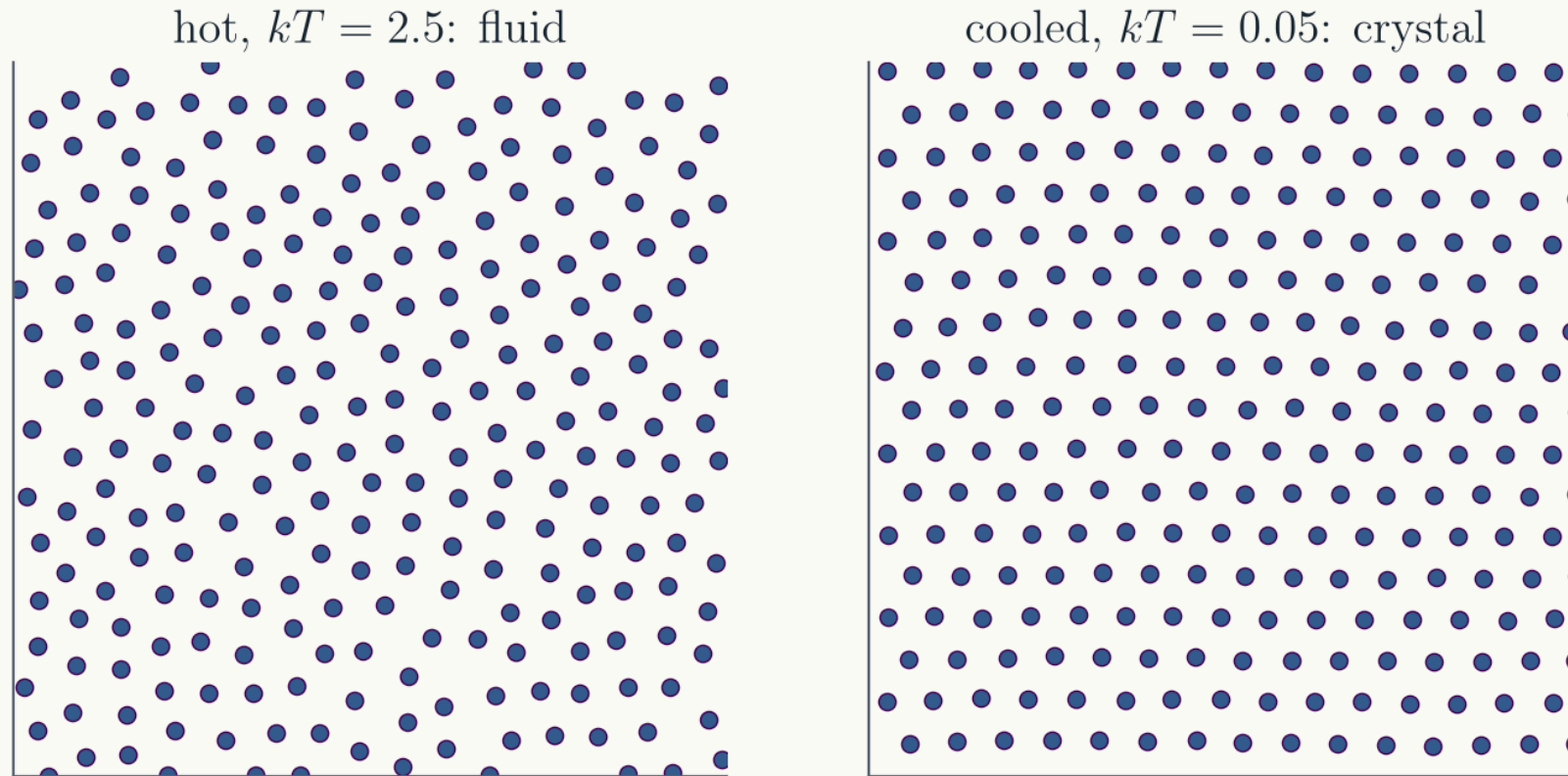


Run 100 atoms for a time  $T$ , reverse every velocity, run  $T$  again. Leapfrog is exactly reversible (Lecture 10), so the gas should return to where it started.

At  $T = 1$  it does, to  $10^{-14}$ . But every 0.35 time units the miss grows by a factor  $e$ : rounding errors are amplified by the chaos of the collisions.

By  $T \approx 12$  the gas misses by half an atom, and after that it keeps no memory of its start.

# Cooling into a crystal



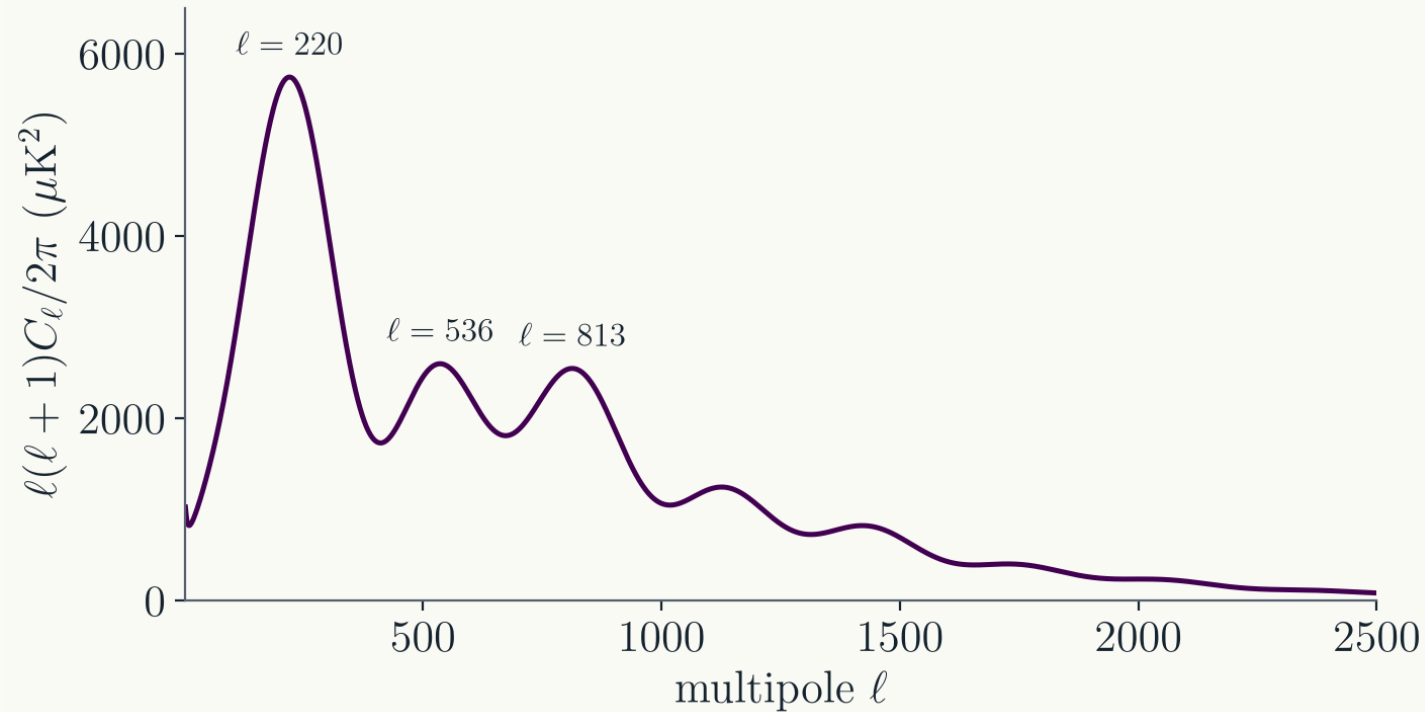
- 256 atoms, slowly cooled by rescaling their velocities every 50 steps.
- The atoms find the triangular lattice by themselves: freezing is in Newton's equations too.

# Part IV · Cosmology



The universe as an ODE problem

# The microwave background



- Temperature fluctuations of about  $10^{-5}$ , left from when the universe became transparent, 380,000 years after the big bang.
- The peaks are sound waves in the early plasma. Their positions and heights measure the contents of the universe.

# Why it is an ODE problem



- On large scales the universe is homogeneous: the background (expansion, density, temperature) depends on time alone.
- The fluctuations are small, so their equations are linear. Fourier transform in space: each wavenumber  $k$  evolves on its own.
- For each  $k$ , every unknown is a function of one variable:

**conformal time**  $\tau$ , with  $d\tau = dt/a(t)$ . A PDE in space and time becomes a set of ODEs in  $\tau$ , one set per  $k$ .

# The ODEs



- **Background:** the Friedmann equation  $\left(\frac{1}{a} \frac{da}{d\tau}\right)^2 = \frac{8\pi G}{3} a^2 \rho$ , for the expansion  $a(\tau)$ .
- **Recombination:** an ODE for the free-electron fraction  $x_e(\tau)$  as protons and electrons form hydrogen.
- **Perturbations, for each  $k$ :** density and velocity of dark matter and of baryons; the photon and neutrino distributions, expanded in multipoles  $\ell$ ; and the metric potentials. For the photons,

$$\frac{dF_\ell}{d\tau} = \frac{k}{2\ell + 1} \left[ \ell F_{\ell-1} - (\ell + 1) F_{\ell+1} \right] - a n_e \sigma_T F_\ell \quad (\ell \geq 3)$$

- **Stiff:** early on, the scattering rate  $a n_e \sigma_T$  is enormous compared with the expansion rate. Codes use a tight-coupling approximation there, or an implicit solver.

# Every pixel is an ODE



## The Doubly Warped World of Binary Black Holes

NASA Goddard



Watch on

NASA Goddard, "The Doubly Warped World of Binary Black Holes": [youtube.com/watch?v=rQcKIN9vj3U](https://www.youtube.com/watch?v=rQcKIN9vj3U)

# References



- Further reading:
  - Qin et al., "Why is Boris algorithm so good?", *Phys. Plasmas* 20, 084503 (2013).
  - Berry & Healy, "Implementation of Gauss-Jackson integration for orbit propagation", *J. Astronaut. Sci.* 52, 331 (2004).
  - Frenkel & Smit, *Understanding Molecular Simulation*.
  - Ma & Bertschinger, *ApJ* 455, 7 (1995); Dodelson & Schmidt, *Modern Cosmology*.